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Glycolysis of Poly(ethylene terephthalate) over Mg–Al Mixed Oxides Catalysts Derived from Hydrotalcites

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ABSTRACT: Poly(ethylene terephthalate) (PET) was depolymerized by ethylene glycol (EG) in the presence of Mg–Al hydrotalcites and their corresponding mixed oxides as solid base catalysts. Mg–Al hydrotalcites with different Mg/Al molar ratios were prepared; it was confirmed by powder X-ray diffraction (XRD) that the materials had hydrotalcite structure. Compared with their corresponding precursors, Mg–Al mixed oxides obtained by the calcination of hydrotalcites exhibited higher catalytic activity for the glycolysis of PET. Furthermore, Mg–Al mixed oxides calcinated at 500 °C with Mg/Al molar ratio of 3 offered the highest catalytic activity for the glycolysis of PET. The experimental results showed that the basicity of catalyst played an important role on the glycolysis activity. The solid catalyst could be easily separated and reused after calcinations again. The influences of experimental parameters on the yield of bis(hydroxyethyl terephthalate)(BHET) were investigated. The evolution of the glycolysis of PET was described by infrared spectroscopy (IR), viscosity-average molecular weight, and scanning electron microscope (SEM).

■ INTRODUCTION

Polyethylene terephthalate (PET) has been extensively used as fibers, film, food packaging, and especially soft drink bottles, and it enjoys such wide applications because of its excellent thermal and mechanical properties. Meanwhile, a large amount of postconsumer PET has been produced. Postconsumer PET does not directly create hazard to the environment, but the disposal of postconsumer PET has caused serious environmental problems because it does not decompose in nature. Furthermore, the recycling of postconsumer PET does also contribute to the conservation of raw petrochemical products and energy. So the effective recycling of PET wastes has attracted a lot of attention over the past decade. Chemical decomposition of PET wastes and conversion into chemical products with high added value, such as unsaturated polyester,[1,2] alkyd resins,[3] and textile softener,[4] is one of the important recycling processes. Chemical depolymerization processes are divided as glycolysis,[5] hydrolysis,[6] methanolysis,[7] and aminolysis.[8] Among the above recycling techniques, glycolysis is regarded as a transesterification process between diols, usually using ethylene glycol(EG), and ester groups of PET to obtain the monomer bis(hydroxyethyl terephthalate)-(BHET).

In previous research, glycolysis of PET is often catalyzed by metal acetate (especially zinc acetate);[5,9] however, it is difficult to separate catalysts from the reaction products due to excellent solubility of metal acetate in EG, and this brings disadvantageous effects on quality of the reaction products because of the toxic nature of heavy metal salts. In an attempt to find an alternative to metal acetate, several eco-friendly catalysts have been investigated for glycolysis reaction, such as titanium phosphate[10] and ionic liquid.[11,12] Troev et al.[10] have pointed out that titanium phosphate showed higher catalytic activity than zinc acetate for glycolysis of PET. Wang et al.[11,12] carried out the depolymerization of PET under different kinds of ionic liquids as catalysts, and it was demonstrated that the purification process of the products in the glycolysis catalyzed by ionic liquids was simpler than that catalyzed by metal acetate. The research[13,14] revealed that alkali catalysts, such as sodium carbonate and sodium bicarbonate, exhibited the catalytic activity for the glycolysis of PET. Recently solid heterogeneous catalysts are preferred over homogeneous ones because of their ease of handling, storage, separation, and disposal. Therefore, it is desirable to find more efficient and cheap heterogeneous catalysts for the glycolysis of PET. More recent three series of solid catalysts including SO_4^{2-}/ZnO, SO_4^{2-}/TiO_2 and $SO_4^{2-}/ZnO\text{-}TiO_2$ are investigated for the glycolysis of PET, the $SO_4^{2-}/ZnO\text{-}TiO_2$ exhibits a catalytic activity with 100% conversion of PET, and 72% selectivity of BHET after 3 h at 180 °C. Moreover, the solid catalyst could be easily separated and reused for four times.[15]

Among various solid catalysts studied in literatures, hydrotalcites are receiving increasing attention. In brief, hydrotalcites are referred to as layered double hydroxide with the general formula $[M_{1-x}^{II}M_x^{III}\ (OH)_2]^{x+}[A^{n-}]_{x/n}\cdot mH_2O$, where M^{II} and M^{III} stand for a divalent and a trivalent cation, respectively, and A^{n-} is the interlayer anion. Hydrotalcites as well as mixed metal oxides formed by calcination of hydrotalcites have been studied as base catalysts in many chemical processes including aldol and Knoevenagel condensations, Michael reactions, and transesterification reactions.[16−19]

In the present work, calcined Mg–Al hydrotalcites are adopted as catalyst for glycolysis PET to BHET monomer; the

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catalytic efficiency is studied regarding the yield of BHET monomer. Further investigation obtained the optimal composition of catalyst and calcination temperature. Moreover, we test the influences of the glycolysis reaction parameters under calcined Mg−Al hydrotalcites as catalysts.

■ EXPERIMENTAL SECTION

2.1. Preparation and Characterization of Hydrotalcites. Mg−Al hydrotalcites were obtained according to the coprecipitation process. In this method, two solutions A and B were simultaneously added dropwise into a flask under vigorous stirring at room temperature; solution A was prepared by mixing the appropriate amounts of Mg and Al metal nitrates in 200 mL of distilled water and solution B was prepared by dissolving 0.35 mol sodium hydroxide and 0.15 mol sodium carbonate in 200 mL of distilled water. The final pH was adjusted to 10. The samples were aged in mother liquor at 65 °C for 1 h. The precipitated solid was filtered, washed with deionized water, and subsequently dried at 50 °C in air to yield hydrotalcites. For convenience, the hydrotalcites thus obtained with different Mg/Al molar ratios of 2, 3, and 4 were designated as HT-2, HT-3 and HT-4. Part of the samples was calcined at different temperatures for 4 h in a muffle furnace to obtain the corresponding mixed oxides. The samples were called CHT-2, CHT-3, and CHT-4. MgO and Al_2O_3 were prepared following the same process as the preparation of CHTs in the absence of the second element.

X-ray diffraction (XRD) patterns of the samples were obtained by Bruker D8 Advance, using filtered CuKα radiation. The metal compositions of the hydrotalcites were determined by inductively couple plasma (ICP) emission spectrometry (IRIS Advantage Radial, ThemoElemental). The specific surface area of the samples was measured by nitrogen adsorption at −196 °C using surface area and pore size analyzers (Autosorb-1-MP, Quantachrome), the samples were degassed at 200 °C for 2 h prior to measurements. The calcined hydrotalcites of 0.15 g were immersed in a toluene solution of phenolphthalein (pK_{BH+} 8.0−9.6) and stirred for 30 min, then titrated with a toluene solution of benzoic acid (0.02 M) to determine the total basicity.[20] Thermal decomposition of the hydrotalcites was evaluated by thermogravimetric analysis (TG) and differential thermal analysis (DTA) carried out on Diamond TG/DTA Instruments under air atmosphere at 10 °C/min from room temperature up to 800 °C.

2.2. Glycolysis of PET and Characterization of Glycolysis Products. Consumed PET soft drink bottles were cut into 2 mm × 2 mm particles for glycolysis experiments after washing and removing caps, labels, and bottom parts. All the reagents were purchased from Sinopharm Chemical Reagent Factory and used without further purification.

A 250 mL round-bottom three necked flask equipped with an electromagnetic stirrer, thermometer and reflux condenser was loaded with 5 g of PET and a certain amount of EG and catalysts. The glycolysis reaction was carried at 160−196 °C in a nitrogen atmosphere. At the end of the reaction, boiling water was added to the round-bottom flask, and the mixture was separated into solid and liquid phases by filtration. The solid product possessed undepolymerized PET, insoluble oligomers, and solid catalyst, and was extracted with boiling water; the water was mixed with the liquid fraction. The white crystalline of BHET was obtained from the filtrate by chilling it. The BHET monomer was purified by repeated crystallization from water, dried in an oven at 50 °C, and weighed to estimate the yield of BHET. The yield of BHET monomer was calculated using the following equations:

$$\text{yield of BHET}(y\%) = \frac{\text{wt of BHET monomer}/M_{\text{BHET}}}{\text{wt of initial PET}/M_{\text{PET}}}100 \tag{1}$$

Where M_{BHET} and M_{PET} refer to the molecular weights of BHET (254g/mol) and repeating unit of PET (192 g/mol), respectively.

^{1}H nuclear magnetic resonance (NMR) was used to verify the chemical structure of the BHET monomer. ^{1}H NMR spectra was recorded with Bruker Avance DPX400 operating at 400 MHz, the spectra was obtained by d6-acetone solution. The melting peaks of BHET from DSC was recorded on STA 409 PC Luxx apparatus with the heating rate of 10 °C/min from 25 to 250 °C under nitrogen atmosphere. Bis(2-hydroxyethyl terephthalate)(BHET): mp, 109 °C.

The glycolysis processes were studied by various characterization techniques. A FTIR spectrum was recorded in a Bruker Equinox 55 instrument, using KBr pellets. The viscosity average molecular weight of PET samples was performed using an Ubbelohde viscometer at 25 °C, PET samples were dissolved in the mixture solution (60 wt % phenol/40 wt % tetrachloroethane). The viscosity average molecular weight of PET samples was calculated from the equation $M = 3.61 \times 10^4 [\eta]^{1.46}$.[5] The morphology of the PET samples was observed on field emission scanning electron instruments (S-4800, Japan).

■ RESULTS AND DISCUSS

Characterization and Selection of Catalysts. Synthesized catalysts are initially characterized by X-ray diffraction (XRD) for their phase purity (Figure 1a). The XRD patterns of HT show sharp, symmetric reflections at low diffraction angles and broad, asymmetric reflections at high diffraction angles, this is typical characteristic of layered materials, such as hydrotalcites. Meanwhile, diffraction profiles of HTs are in good agreement with earlier measurement in previous literatures.[19,20] When heating the HTs up to 500 °C, the calcinations destroy the layered structure. CHT samples exhibit the typical feature of MgO-like phase as identified their XRD spectra ($2\theta \approx 43$ and 63°) (Figure 1b) and the reflections are broadened due to poor crystallization or small particle size; it is suggested that Al^{3+} cation were dispersed in the structure of MgO.[20,21]

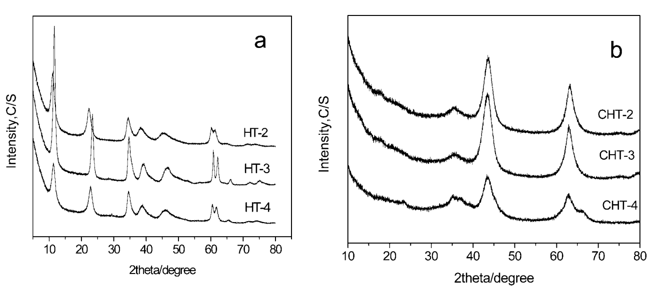


Figure 1. XRD patterns of hydrotalcite samples. (a) Uncalcined samples with various Mg/Al molar ratios; (b) samples with various Mg/Al molar ratios at the calcination temperature of 500 °C.

The thermostability of Mg/Al hydrotalcite is investigated by the TG/DTA method. For the sample of HT-3 (in Figure 2), gradual weight loss is observed from about 60 to 600 °C, with two main endothermic effects at about 211 and 392 °C. As shown in Figure 2, the first endothermic effect below 242 °C,

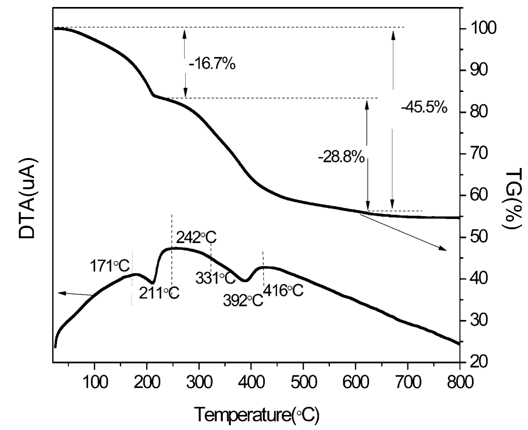


Figure 2. TG-DTA curves of HT-3 sample.

which is accompanied by a mass loss 16.7%, involves the elimination of the surface weakly bound water and interlayer water of Mg/Al hydrotalcite. The hydrotalcite structure will be retained in this temperature range. The second endothermic DTA peak at 392 °C, mass loss of approximately 28.8%, is attributed to the decomposition of the carbonate anion in the brucite-like layers, and the deeper decomposition of brucite layer OH^- anions.

Analytical data of metals in the samples (see Table 1) indicate that the Mg/Al molar ratios in the final products are similar to those in the initial synthesis mixtures, the small difference is due to the favored precipitation of aluminum hydroxide because of its higher pK_a value.[22] The BET specific surface area for the hydrotalcites and calcined samples are in accordance with those reported in the literature.[23] The surfaces of the hydrotalcites were increased during the calcination process, it should be supposed that CO_2 evolved from carbonate upon the thermal decomposition of hydrotalcites and developed a porous system of the calcined samples.

Table 1. Physicochemical Properties and Catalytic Activities of HTs and CHTs with Different Mg/Al Molar Ratios

catalyst	measured Mg/Al molar ratio[a]	S_{BET} (m^2/g)	basicity (mmol/g)[b]	yield (%)[c]
HT-2	1.89	53.4		61.4
HT-3	2.83	50.1		66.4
HT-4	3.77	66.3		31.3
Al_2O_3		251.4	0.11	10.7
CHT-2[b]		171.3	0.21	76.2
CHT-3[b]		182.1	0.34	81.3
CHT-4[b]		187.6	0.20	65.6
MgO		36.2	0.39	69.3

[a]Determined by ICP. [b]Calcination temperature: 500 °C. [c]Reaction conditions: the weight ratio of catalyst to PET is 1%; the weight ratio of EG to PET is 5; reaction time, 50 min; reaction temperature, 196 °C.

The surface of the CHTs contain basic sites of low (OH^- groups), medium (Mg–O pairs), and strong (O^{2-}) basicities.[24,25] The basicity of the CHTs is determined by using Hammett indicator, and the basicity of the CHTs with different Mg/Al molar ratios and calcination temperatures are shown in Table 1 and Table 2. The experimental results show that Mg/Al

Table 2. Catalytic Activities of CHT-3 Calcined at Different Temperatures

catalysts (calcination temperature)	basicity (mmol/g)	yield (%)[a]
CHT-3(300 °C)	0.14	54.2
CHT-3(400 °C)	0.29	79.1
CHT-3(500 °C)	0.34	81.3
CHT-3(600 °C)	0.26	70.7
CHT-3(800 °C)	0.19	61.4

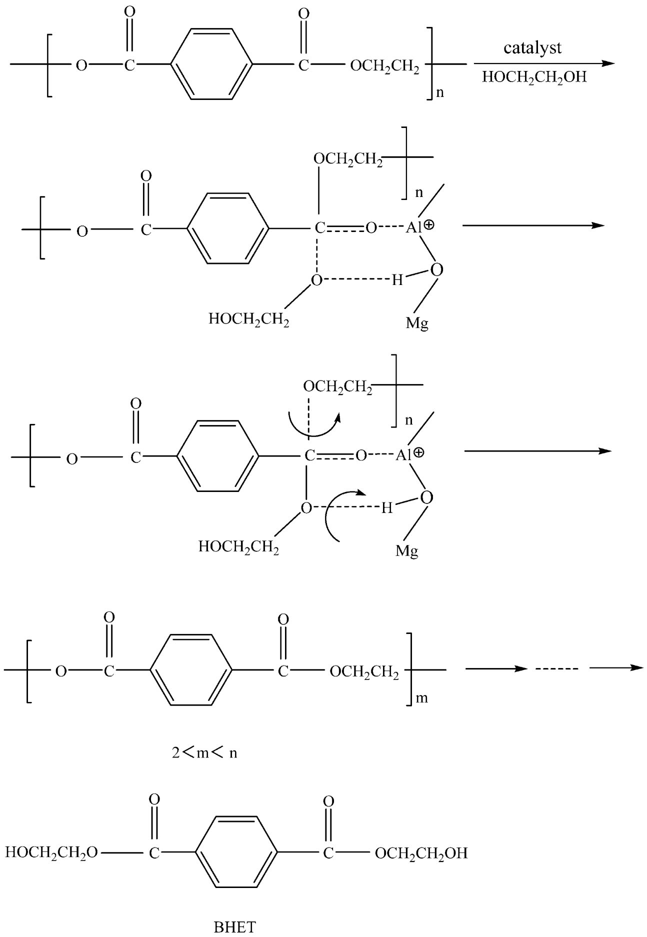
[a]Reaction conditions: the weight ratio of catalyst to PET is 1%; the weight ratio of EG to PET is 5; reaction time, 50 min; reaction temperature, 196 °C.

molar ratios and calcination temperatures have an influence on the basicity of the CHTs. The samples with Mg/Al molar ratios of 3 have the highest basicity; similar results are also reported by other researchers.[20,24] Furthermore, the basicity of CHT-3 calcined at different temperatures is measured, it is observed that the basicity of CHT-3 is increased with an increase of calcination temperature and reached 0.34 mmol/g at 500 °C, but the basicity is decreased with further increasing calcination temperature.

The catalytic activity of HTs and CHTs are investigated, the experimental results show that CHTs display significantly higher catalytic activity in the glycolysis of PET, compared with the corresponding HTs. Regarding the pure oxides, pure Al_2O_3 does not display appreciable catalytic activity, and this may be related to its weak basicity or acidity. When Al^{3+} is added into the MgO crystal, the defect of Mg^{2+} or Al^{3+} is produced in order to compensate the positive charge generated. The O^{2-} ions adjacent to the Mg^{2+} or Al^{3+} defects become coordinatively unsaturated and can provide strong basic sites. In other words, it is known that incorporated Al is more electronegative than Mg, thus the average electronic density of the unsaturated framework oxygens is increased with increasing Mg/Al ratio, resulting in an increase in their basicity and catalytic activity. When the Mg/Al ratio exceeds 3, the basicity and catalytic activity decreases, which may be due to the formation of new weaker basic sites and thus a decrease of strong basic site amount.[24] As can be seen, the results obtained with CHTs in the range $2 \leq Mg/Al \leq 4$ show the maximum yield of BHET is obtained when the Mg/Al molar ratios is 3.

The MgO sample does not show acid sites, but calcined MgAl hydrotalcites can be viewed as possessing a pair of strong basic sites and weak acid sites; acid sites are generated when Al is added into MgO lattice.[25] The experimental results also show that the basicity of catalyst plays an important role on the glycolysis activity, but the weak acidity of a catalyst can help to improve the catalytic activity. We propose a possible catalytic mechanism of the glycolysis reaction under CHTs as shown in Scheme 1. The hydrogen in the hydroxyl group of ethylene glycol interacts with the basic sites on the surface of the catalyst, Meanwhile, the carbonyl oxygen (C═O) in the ester combines with the neighboring acidic sites of catalyst, which leads to the carbon cations becoming more electropositive. Then oxygen in the hydroxyl group of the ethylene glycol nucleophilic attacks the carbon cation to form a tetrahedral intermediate. In the end, the tetrahedral intermediate is rearranged, these processes repeat, and BHET monomer is formed. Therefore, MgO is not able to promote the glycolysis reactions at the experimental conditions in spite of its high density of strong basic sites. The small surface area of pure MgO may also be the factor to inhibit its catalytic activity on glycolysis of PET (in Table 1).

Scheme 1. Mechanism of the Glycolysis of PET over Mg−Al Mixed Oxides Catalysts Derived from Hydrotalcites



The influence of calcination temperature on the catalyst activity is displayed in Table 2. As expected, the maximum yield of BHET reaches to 81.3% in the reaction time of 50 min with CHT-3 calcined at 500 °C, and then it is found that the yield of BHET decreases with a further increase in calcination temperatures, which lies in a drop of the basicity of the catalyst. Therefore, CHT-3 derived from HT-3 calcined at 500 °C is chosen as the glycolysis catalyst for further investigation.

Influences of Reaction Conditions. We investigate the effect of amount of CHT-3 catalyst on the yield of BHET. The experiments are carried out with the weight ratio (EG to PET) of 5 at 196 °C in the reaction time of 50 min. The results are summarized in Figure 3. It can be observed that the increase of the weight ratio of CHT-3 to PET from 0.5 to 1% causes a dramatic increase of yield of BHET from 41.7% up to 81.3%. When the weight ratio of CHT-3 to PET is 1%, the yield of BHET reaches a high level and then increases slightly.

Five experiments are carried out varying the weight ratio of EG to PET between 3:1 and 7:1. Figure 4 presents the relationship between yield of BHET and EG to PET ratio. As it can be observed, with the weight ratio (EG to PET) of 3, the yield of BHET is near 68.1% after 50 min. An increase of the weight ratio from 3 to 5 causes significant increase of the yield of BHET from 68.1% to 81.3%. When the weight ratio of EG to PET is 5, the yield of BHET reaches high level and then increases slowly. Taking into account increases the weight ratio of EG to PET demands larger capital cost and energy consumption. On the basis of experimental results, the optimal parameter of weight ratio of EG to PET is 5.

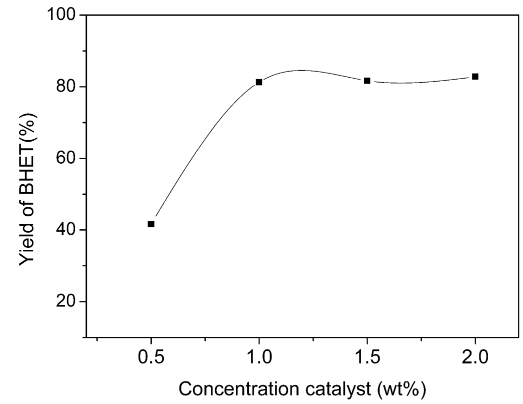


Figure 3. The effect of the weight ratio of catalyst to PET on the yield of BHET (the weight ratio of EG to PET of 5, 196 °C, 50 min).

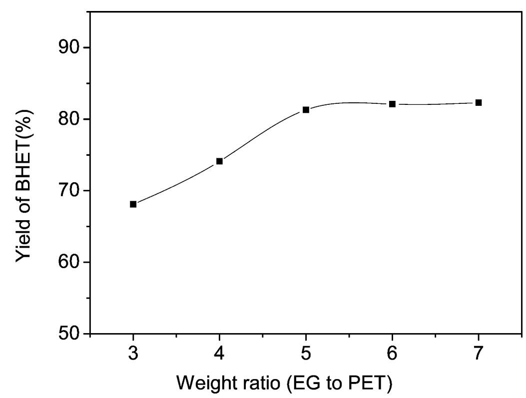


Figure 4. The effect of the weight ratio EG to PET on the yield of BHET (the weight ratio of catalyst to PET of 1%, 196 °C, 50 min).

Figure 5 shows the influence of reaction temperature in the range of 160−196 °C on the yield of BHET at the weight ratio (EG to PET) of 5, the weight ratio (CHT-3 to PET) of 1% and reaction time of 50 min. The results show that the yield of BHET is close to 0% at 160 °C, and then the yield of BHET increases with increasing reaction temperature because the glycolysis reaction of PET with EG is endothermic.[26]

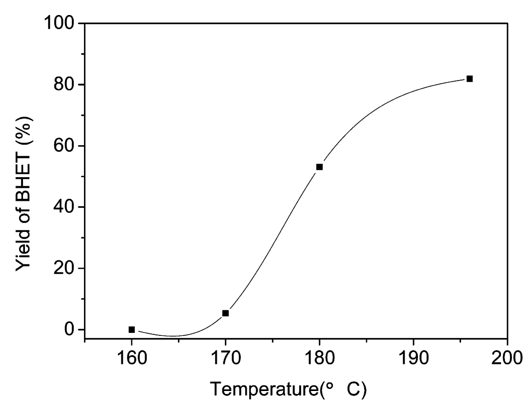


Figure 5. The effect of reaction temperature on the yield of BHET (the weight ratio EG to PET of 5, the weight ratio of catalyst to PET of 1%, 50 min).

Moreover, the proportion of BHET monomer in the products increases with an increase in the reaction temperature.[11,26]

The effects of glycolysis time on the yield of BHET are presented in Figure 6. The experiments are carried out with the weight ratio (EG to PET) of 5 and the weight ratio (CHT-3 to PET) of 1% at 196 °C. As can be seen in Figure 6, within the first 50 min, the yield of BHET increased rapidly with the extension of the glycolysis reactions, and after 50 min, the yield of BHET increased slightly. The glycolysis of PET is a reversible equilibrium reaction, and the reverse reaction is polycondensation. The experimental results indicate that the equilibrium state is reached after 50 min.

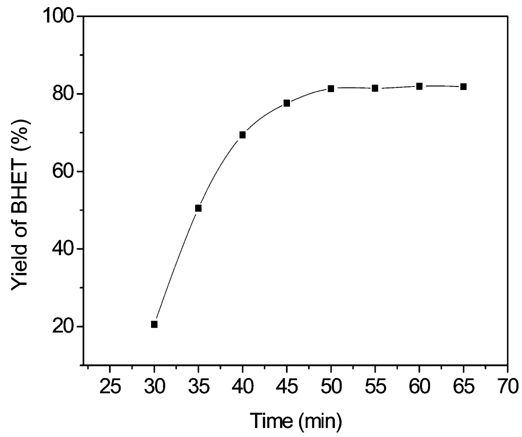


Figure 6. The effect of reaction time on the yield of BHET (the weight ratio EG to PET of 5, the weight ratio of catalyst to PET of 1%, 196 °C).

Catalyst Deactivation and Regeneration. The reusability of heterogeneous catalysts is a key factor in practical applications, so the reusability of the catalyst is examined. After glycolysis reactions, solid catalyst, along with unreacted PET, remains in the filter residue. To investigate the reusability of catalyst, solid catalyst must be separated from unreacted PET. In this study, the filter residues are impregnated into the mixture solution (60 wt % phenol/40 wt % tetrachloroethane), where unreacted PET dissolves, and then the solid catalyst is obtained by filtration.

Due to "memory effect",[27] the mixed oxide should be recovered to the partial layered structure of hydrotalcite in the presence of water during the the filtration procedure of the glycolysis reaction. This fact is confirmed by the XRD patterns of used CHT-3, which is shown in Figure 7 (curve a). Results show that the yield of BHET decreases sharply to 8.9% under the optimal parameter of glycolysis reactions, when the catalyst is reused without any after-treatment. The metal elemental composition of used CHT-3 is determined by inductively coupled plasma and listed in Table 3. From the analysis results, it can be found that the atom ratio of Mg to Al remained almost unchanged, which meant that leaching of the Mg and Al element is not the primary reason that caused the deactivation of catalysts. However, the basicity of used CHT-3 sharply decreased, which led to the degradation of catalysis performance. The decay of basicity may be associated with degradation of the strong Lewis base sites (O^{2-}) during the recycling processes in the presence of water.[27] The recovered catalyst is approximately 85% through filtration, solution, and washing processes. Meanwhile, we also investigate the content of Mg and Al element of the BHET; the experimental results show that negligible amounts of catalyst leach in the glycolysis product (Mg, 40 ppm; Al, 37 ppm). This suggests that the glycolysis reaction is actually a heterogeneous catalytic process in the presence of mixed oxides as catalyst.

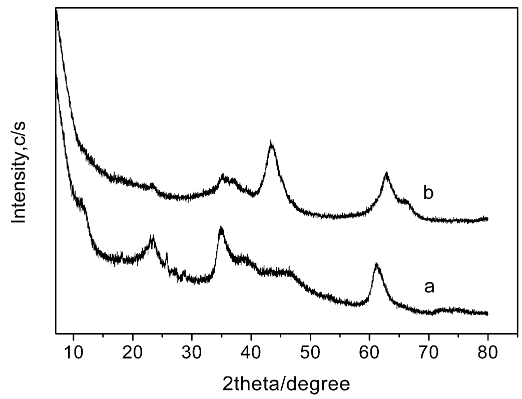


Figure 7. XRD patterns of used catalyst (a) and treated catalyst (b).

Table 3. The Mg/Al Molar Ratio and Basicity of the CHT-3 before and after Usage

catalyst	measured Mg/Al molar ratio	basicity (mmol/g)
CHT-3	2.83	0.34
CHT-3(used)	2.78	0.05

On the other hand, used mixed oxides are calcined at 500 °C again; the characteristic diffraction peak of used mixed oxide is similar to the fresh mixed oxide, but one has a relatively low peak (Figure 7 curve b). Therefore we perform four consecutive glycolysis reactions using mixed oxide during the regeneration process. Results show that the catalyst could be recycled at least three times under the optimal parameter of glycolysis reactions although the catalytic activity decreased with the reuse of catalyst (first, 81.3%; second, 64.2%; third, 62.7%; fourth, 38.8%). These results point to recalcination as a simple and reliable way to regenerate catalyst.[27]

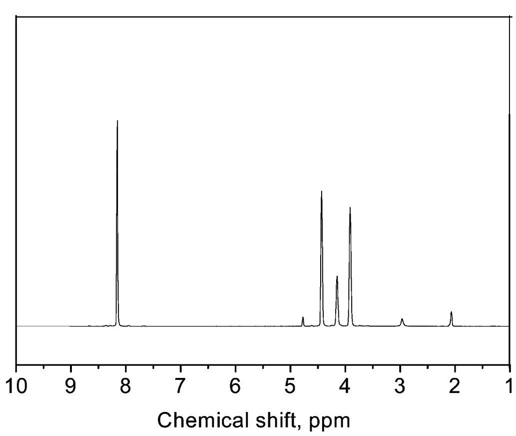


Figure 8. ^1H NMR spectra of BHET monomer.

Evolution of the Glycolysis of PET. Figure 8 shows the ^1H NMR spectrum of the BHET. The signal at δ 8.1 ppm shows that the presence of the four aromatic protons of the benzene ring, signal at δ 3.9 ppm is characteristic of the methylene protons of CH_2−OH, the signal at δ 4.4 ppm indicates the protons of the methylene group near −COO−, the signal at δ 4.1 ppm is attributed to the protons of the hydroxyl group. The ^{1}H NMR data resemble well those of reported data in the literature.[13] The melting point of the glycolysis product displays at 109 °C in agreement with the known melting point of BHET,[9,13] this is evidence of the high level of purity of the BHET monomer produced by our process.

The IR spectra of the BHET monomer and residual PET are shown in Figure 9. The IR spectra shows the −OH band at

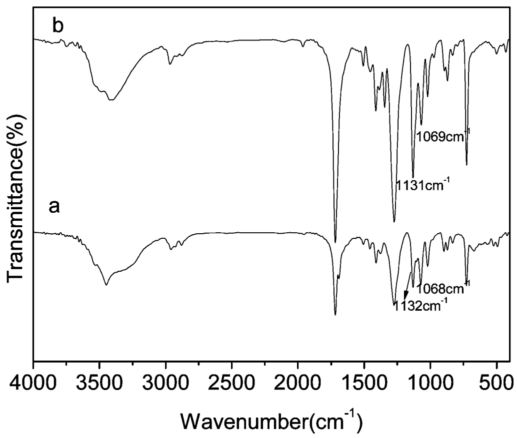


Figure 9. IR spectra of BHET monomer and residual PET: (a) BHET monomer, (b) residual PET (the weight ratio EG to PET of 5, the weight ratio of catalyst to PET of 1%, 196 °C, 40 min).

3449 cm^{-1}, alkyl C−H at 2878 and 2954 cm^{-1}, C=O stretching at 1718 cm^{-1}, the ester C−O peak at 1132 and 1273 cm^{-1}, the primary alcohol C−O stretching at 1068 cm^{-1}, and benzene ring peak at 1570 cm^{-1} as well as at 727 and 871 cm^{-1}.[28] It can be seen that the two spectra are similar because no new groups are formed during glycolysis process. But compared with the absorption peak of primary alcohol C−O stretching at 1068 cm^{-1} and the ester C−O peak at 1132 cm^{-1} of residual PET (in Figure 9b), the primary alcohol C−O stretching of the BHET monomer is relatively strong (in Figure 9a). This suggests that ester linkages are broken and replaced with hydroxyl terminals during glycolysis reaction.

To further reveal evolution of the glycolysis of PET, the fresh PET and residual PET under different reaction times are analyzed by scanning electron microscope and viscosity-average molecular weight.

Figure 10 shows the surface of the residual PET particle at different reaction times of glycolysis. Figure 10a shows that the surface of PET is relatively smooth at the reaction time of 20 min. In this reaction stage, the glycolysis of PET occurs at the external area of PET and catalyst. However, the formation of the porous structure is observed at the reaction time of 30 min (in Figure 10b). Figure 10 panels c and d indicate that the surface of PET become significantly more microporous. Formation of a porous structure not only increases the effective reaction area but also promotes penetration of EG and catalyst into a solid PET particle. Therefore the rate of glycolysis of PET should be increased from 30 min; this is in agreement with the experimental results (in Figure 6). On the other hand, the rate of concentration of the solid oligomer at the surface of the PET increases as the reaction goes on. The two effects can be carried out simultaneously so that the rate of glycolysis of PET may be constant after a reaction time of 50 min.

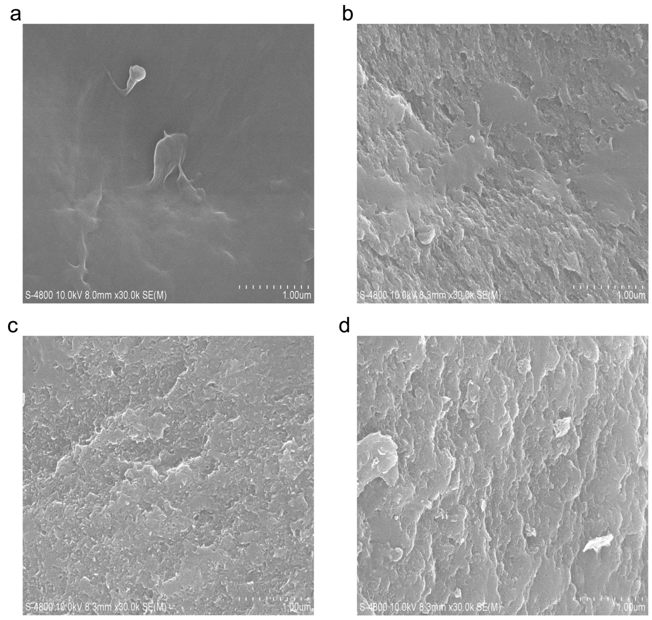


Figure 10. SEM photographs of the residual PET: (a) after 20 min; (b) after 30 min; (c) after 35 min; (d) after 45 min.

The viscosity-average molecular weight of the fresh and residual PET was carried out, and the results are presented in Table 4. The table shows that the viscosity-average molecular

Table 4. Viscosity-Average Molecular Weight of Fresh and Residual PET under Different Reaction Time

reaction time(min)	viscosity-average molecular weight
0	25880
20	19840
30	10630
35	2350
40	704

weight of the fresh PET is 25 800, and increasing reaction time can decrease the viscosity-average molecular weight of the residual PET. This occurs because, for residual PET with a porous structure, EG and a small particle size catalyst may penetrate into a PET particle and break the chains of PET, which results in the decrease of its molecular weight. In other words, the random scission reaction occurs not only at the surface but also inside of the PET particle in the middle of the glycolysis reaction stage.

■ CONCLUSIONS

Calcined Mg−Al hydrotalcites were found to be effective catalysts for glycolysis of PET. These mixed oxide catalysts were inexpensive and nontoxic, thus it was concluded that calcined Mg−Al hydrotalcites were promising candidates for the glycolysis reactions and would be able to replace homogeneous catalysts. The experimental results showed that calcined Mg−Al hydrotalcites were the most active catalyst when the molar ratio of Mg to Al was 3 and calcined at 500 °C. The catalytic activity correlated with the basicity of catalysis. The yield of BHET increased with increasing amount of catalyst, the weight ratio of EG to PET, glycolysis temperature, and time until the glycolysis reaction reached equilibrium states. Optimum reaction conditions were obtained with the weight ratio (CHT-3 to PET) of 1% and the weight ratio (EG to PET) of 5 for 50 min reaction at 196 °C. The calcined Mg−Al hydrotalcites could be separated and reused at least three times.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Pimpan, V.; Sirisook, R.; Chuayjuljit, S. Synthesis of unsaturated polyester resin from post consumer PET bottles: Effect of type of glycol on characteristics of unsaturated polyester resin. *J. Appl. Polym. Sci.* **2003**, *88*, 788.

(2) Nikles, D. E.; Farahat, M. S. New motivation for the depolymerization products derived from poly(ethylene terephthalate) (PET) waste: A review. *Macromol. Mater. Eng.* **2005**, *290*, 13.

(3) Karayannidis, G. P.; Achilias, D. S.; Sideridou, I. D; Bikiaris, D. N. Alkyd resins derived from glycolized waste poly(ethylene terephthalate). *Eur. Polym. J.* **2005**, *41*, 201.

(4) Shukla, S. R.; Harad, A. M.; Jawale, L. S. Recycling of waste PET into useful textile auxiliaries. *Waste. Manage.* **2008**, *28*, 51.

(5) Ghaemy, M.; Mossaddegh, K. Depolymerisation of poly(ethylene terephthalate) fibre wastes using ethylene glycol. *Polym. Degrad. Stab.* **2005**, *90*, 570.

(6) Yoshioka, T.; Okayama, N.; Okuwaki, A. Kinetics of hydrolysis of PET powder in nitric acid by a modified shrinking-core model. *Ind. Eng. Chem. Res* . **1998**, *37*, 336.

(7) Kurokawa, H.; Ohshima, M.; Sugiyama, K.; Miura, H. Methanolysis of polyethylene terephthalate (PET) in the presence of aluminium triisopropoxide catalyst to form dimethyl terephthalate and ethylene glycol. *Polym. Degrad. Stab.* **2003**, *79*, 529.

(8) Jain, A.; Soni, R. K. Spectroscopic investigation of end products obtained by ammonolysis of poly(ethylene terephthalate) waste in the presence of zinc acetate as a catalyst. *J. Polym. Res.* **2007**, *14*, 475.

(9) Xi, G. X.; Lu, M. X.; Sun, C. Study on depolymerization of waste polyethylene terephthalate into monomer of bis(2-hydroxyethyl terephthalate). *Polym. Degrad. Stab.* **2005**, *87*, 117.

(10) Troev, K.; Grancharov, G.; Tsevi, R.; Gitsov, I. A novel catalyst for the glycolysis of poly(ethylene terephthalate). *J. Appl. Polym. Sci.* **2003**, *90*, 1148.

(11) Wang, H.; Liu, Y. Q.; Li, Z. X.; Zhang, X. P.; Zhang, S. J.; Zhang, Y. Q. Glycolysis of poly(ethylene terephthalate) catalyzed by ionic liquids. *Eur. Polym. J* . **2009**, *45*, 1535.

(12) Wang, H.; Yan, R. Y.; Li, Z. X.; Zhang, X. P.; Zhang, S. J. Fe-containing magnetic ionic liquid as an effective catalyst for the glycolysis of poly(ethylene terephthalate). *Catal. Comm.* **2010**, *11*, 763.

(13) López-Fonseca, R.; Duque-Ingunza, I.; Rivas, B. D.; Arnaiz, S.; Gutiérrez-Ortiz, J. I. Chemical recycling of post-consumer PET wastes by glycolysis in the presence of metal salts. *Polym. Degrad. Stab.* **2010**, *95*, 1022.

(14) Pingale, N. D.; Shukla, S. R. Microwave assisted ecofriendly recycling of poly(ethylene terephthalate) bottle waste. *Eur. Polym. J.* **2008**, *44*, 4151.

(15) Zhu, M. L.; Li, S.; Li, Z. X.; Lu, X. M.; Zhang, S. J. Investigation of solid catalysts for glycolysis of polyethylene terephthalate. *Chem. Eng. J* **2012**, *185−186*, 168.

(16) Tichit, D.; Lutic, D.; Coq, B.; Durand, R.; Teissier, R. The aldol condensation of acetaldehyde and heptanal on hydrotalcite-type catalysts. *J. Catal.* **2003**, *219*, 167.

(17) Mei, F. M.; Pei, Z.; Li, G, X. The transesterification of dimethyl carbonate with phenol over Mg-Al-hydrotalcite catalyst. *Org. Process Res. Dev.* **2004**, *8*, 372.

(18) Choudary, B. M.; Kantam, M. L.; Reddy, C. R. V.; Rao, K. K.; Figueras, F. The first example of Michael addition catalysed by modified Mg−Al hydrotalcite. *J. Mol. Catal.* **1999**, *146*, 279.

(19) Churchil, A. A.; Kannan, S. Hantzsch pyridine synthesis using hydrotalcites or hydrotalcite-like materials as solid base catalysts. *Appl. Catal., A* **2008**, *338*, 121.

(20) Fraile, J. M.; Garcia, N.; Mayoral, J. A.; Pires, E.; Roldan, L. The influence of alkaline metals on the strong basicity of Mg−Al mixed oxides: The case of transesterification reactions. *Appl. Catal., A* **2009**, *364*, 87.

(21) Zeng, H. Y.; Zhen, F.; Deng, X.; Li, Y. Q. Activation of Mg−Al hydrotalcite catalysts for transesterification of rape oil. *Fuel* **2008**, *87*, 3071.

(22) Chuayplod, P.; Trakarnpruk, W. Transesterification of rice bran oil with methanol catalyzed by Mg(Al)La hydrotalcites and metal/MgAl oxides. *Ind. Eng. Chem. Res* . **1998**, *37*, 336.

(23) Kustrowskia, P.; Chmielarza, L.; Bozeka, E.; Sawalhab, M.; Roessner, F. Acidity and basicity of hydrotalcite derived mixed Mg−Al oxides studied by test reaction of MBOH conversion and temperature programmed desorption of NH_3 and CO_2. *Mater. Res. Bull.* **2004**, *39*, 263.

(24) Xie, W. L.; Peng, H.; Chen, L. G. Calcined Mg−Al hydrotalcites as solid base catalysts for methanolysis of soybean oil. *J. Mol. Catal., A* **2006**, *246*, 24.

(25) Cosimo, J. I. D.; Diez, V. K.; Xu, M.; Iglesia, E.; Apesteguia, C. R. Structure and surface and catalytic properties of Mg-Al basic oxides. *J. Catal.* **1998**, *178*, 499.

(26) Imran, M.; Kim, B. K.; Han, M.; Cho, B. G.; Kim, D. H. Sub- and supercritical glycolysis of polyethylene terephthalate (PET) into the monomer bis(2-hydroxyethyl) terephthalate (BHET). *Polym. Degrad. Stab.* **2010**, *95*, 1686.

(27) Liu, Y. J.; Lotero, E.; Goodwin, J. G., Jr; Mo, X. H. Transesterification of poultry fat with methanol using Mg−Al hydrotalcite derived catalysts. *Appl. Catal., A* **2007**, *331*, 138.

(28) Goje, A. S.; Mishra, S. Chemical kinetics, simulation, and thermodynamics of glycolytic depolymerisation of poly(ethylene terephthalate) waste with catalyst optimization for recycling of value added monomeric products. *Macromol. Mater. Eng.* **2003**, *288*, 326.